Md Kaif Imteyaz

kaifimteyaz.k@gmail.com | kaif-imteyaz.github.io | linkedin.com/kaifimteyaz | scholar.google.com/kaifimteyaz

Summary

Research Engineer with experience building applied AI pipelines for government data and healthcare. Work includes structured extraction from scanned documents at scale, LLM and VLM evaluation, and spatiotemporal deep learning, with ongoing work on AI-assisted dental record digitisation at AIIMS–IIT Delhi–UCL.

Education

Jamia Hamdard — New Delhi, India
Bachelor of Technology in Computer Science & Engineering — 2023 – 2026

Skills

Programming: Python, JavaScript

ML & Deep Learning: PyTorch, Scikit-learn, OpenCV, CNNs, InceptionV3, Attention Mechanisms (CBAM), Transfer Learning, Fine-Tuning, Model Quantization

AI Research: LLMs, VLMs, Hugging Face, Qwen3-VL, OpenAI API, Ollama, RAG, RLHF, LoRA, QLoRA, Explainable AI (XAI), Hallucination Detection, AI Fairness & Bias, Ablation Studies, Multi-Seed Evaluation, Medical Imaging

Data & MLOps: NumPy, Pandas, Matplotlib, Seaborn, Plotly, Weights & Biases (W&B), Hypothesis Testing, Cohort Analysis

Web & Backend: FastAPI, Node.js, Express.js, React.js, Next.js, Tailwind CSS, Supabase

Databases: PostgreSQL, MySQL

Awards

- **Won 2nd Place**, AMD India Reinforcement Learning Hackathon among 200+ teams, organized by AMD, Yardi School of AI (IIT Delhi), and Unsloth AI (2026)
- **7th Rank** (out of 2,000+ participants, two-phase), AISEHack 2026 organized by ANRF in collaboration with IBM and IIT Delhi, hosted at IIIT Hyderabad (2026)

Experience

Research Engineer — Aug. 2026 – present
Working with Dr. Prateek Raj, University College London

- Working on a joint AIIMS–IIT Delhi–UCL research project applying AI and HCI to digitise paper-based dental records and improve accessibility of digital healthcare.

Research Intern (AI & Data Engineering) — Apr. 2026 – Oct. 2026
King's College London

- Built an end-to-end pipeline extracting structured asset and liability data from scanned Bihar government officer affidavits (pilot scope: 4 districts) using a vision-language model (Qwen3-VL), converting raster scans with no embedded text layer into a validated 78-field schema.
- Engineered DOB, PAN, and income-tax field parsing plus source-aware deduplication preserving cross-year filings and tuned 4-bit GPU inference with adaptive batching, cutting extraction time from 285s to 23s per document.
- Delivered dual CSV & JSON outputs with source-PDF links and confidence flags, achieving 98.8% valid DOB and 90.8% valid PAN extraction, with remaining gaps traced to blank or illegible source documents.

Data Science Intern — Sept. 2025 – Feb. 2026
Worked with Prof. Aaditya Dar, Independent Researcher

- Implemented a scalable data pipeline that cleaned, standardized, and structured approximately 2 million government official financial asset declaration records for downstream use.
- Designed and executed comparative experiments across BART, FLAN-T5, GPT-2, and Qwen models, evaluating performance trade-offs for structured information extraction tasks.
- Evaluated encoder architectures for entity standardization and record linkage, analyzing trade-offs in accuracy, generalization, and inference efficiency.

Research Intern — Mar. 2025 – Aug. 2025
Worked with Prof. Prabhat Barnwal, Michigan State University

- Developed the Peer Knowledge Exchange Platform to improve retrieval speed, accessibility, and user experience for government officers in Punjab, India.
- Engineered Sewa Assist, a conversational retrieval interface deployed via WhatsApp, enabling officers to access program resources through natural language queries.
- Integrated an on-site chatbot that reduced user friction and improved resource discovery through keyword-driven retrieval logic.

Full Stack Developer Intern Dec. 2022 – Jan. 2023
FTK Center for Information Technology, Jamia Millia Islamia New Delhi, India

- Led full stack development of an alumni community web application serving 2,000+ users, improving engagement using HTML, CSS, JavaScript, and PHP.
- Optimized site architecture for mobile responsiveness and reduced page load times by 20%.
- Collaborated in a four-member Agile team to ship three major features within six weeks.

Research Projects

Design Theater: A Benchmark for Generative UI (Accepted at AIES 2026) | LLM Benchmarking

Developed a benchmark to measure authenticity gaps between generated reasoning traces and actual model behavior through systematic evaluation and interpretability analysis.

Episode-Aware PM2.5 Forecasting Across India | Spatiotemporal Deep Learning

Designed a spatiotemporal deep learning model for country-level PM2.5 forecasting over a 140×124 grid at 25 km resolution, predicting 16 future hours from a 10-hour lookback across 26 input channels with no future meteorological context.

CATS: Culturally Adaptive Therapeutic System | Final Undergraduate Project

Designed an agentic system for physiotherapists to generate personalized exercise videos aligned with patients' cultural backgrounds through culturally appropriate demonstrations.

Projects

Ink2Info: Digitizing Health Informatics | *Python, Next.js, FastAPI, OpenAI, Gemma 7B* | Repository 2026

- Developed a prescription digitization pipeline using OpenAI and Gemma 7B with a Fabric.js canvas for real-time handwriting extraction.
- Implemented role-based access control with HMAC-signed QR codes for secure data sharing across multiple user roles.

De-Tail: Agentic AI Search Assistant | *React, OpenAI API, Tavily API* | Repository 2025

- Engineered a retrieval-augmented generation pipeline that exposes its full step-by-step reasoning chain, integrating OpenAI Reasoner with Tavily API for real-time web retrieval.
- Designed end-to-end retrieval flow from query intake through document fetching, reranking signal generation, and response synthesis, with full transparency at each stage.

Certifications & Activities

- **IBM SkillsBuild Certifications:** Artificial Intelligence, Data Fundamentals
- **AI Summer School**, IIIT Delhi (May 2024, July 2023)